

Neural communication

Neurons communicate through electrical and chemical synapses

Computational Neuroscience

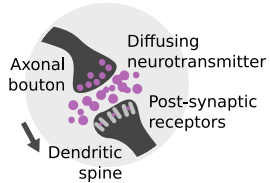
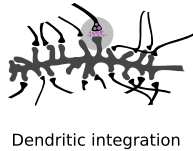
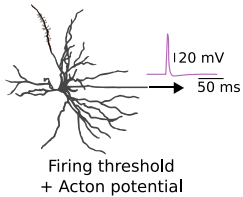
University of Bristol

M Rule

Learning outcomes:

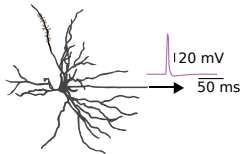
- ▶ What is a synapse?
- ▶ Key terms
- ▶ Steps of synaptic transmission
- ▶ Describe what synapses are and how they work
- ▶ Explain the steps of electrochemical transmission in synapse

Information flow in single neurons



Information flow in single neurons

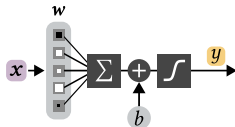
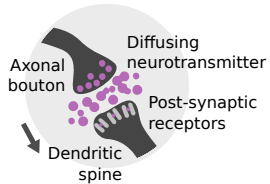
Biological



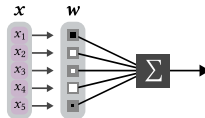
Firing threshold
+ Action potential



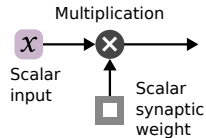
Dendritic integration



Additive bias
+ Saturating nonlinearity

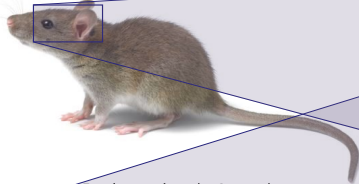


Linear summation



Artificial

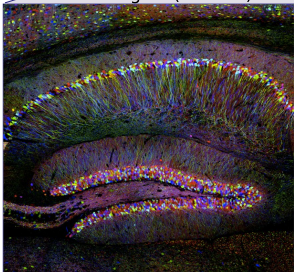
Animal (~10 cm)



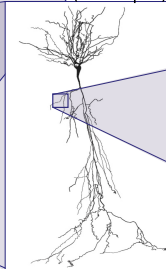
Brain (~1 cm)



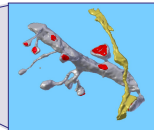
Brain region (~1 mm)



Neuron (~100 μm)

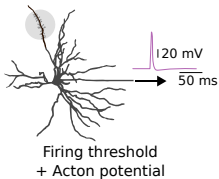
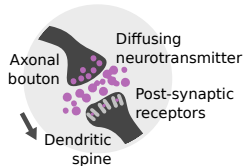


Synapses (~1 μm)



Modified from B. Mizusaki (2022) lecture slides

Neurons communicate through electrical and chemical synapses



Electrical synapses (“gap junctions”) connect the interiors of two neurons.

ask me about Camillo Golgi and the syncytium

Chemical synapses

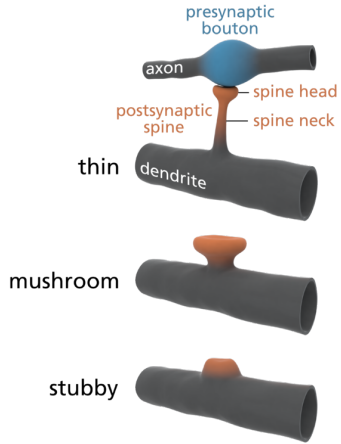
- ▶ $\sim 1 \mu\text{m}$ in size;
- ▶ Δ synaptic strength $\rightarrow$ learning, memory
- ▶ **Presynaptic** spike $\rightarrow$ **Postsynaptic** potential (PSP)
- ▶ Sometimes unreliable
- ▶ May have complicated, time-dependent, nonlinear dynamics (some computation happens here)

Excitatory

- ▶ Make targets more likely to spike
- ▶ **Glutamate** (others)

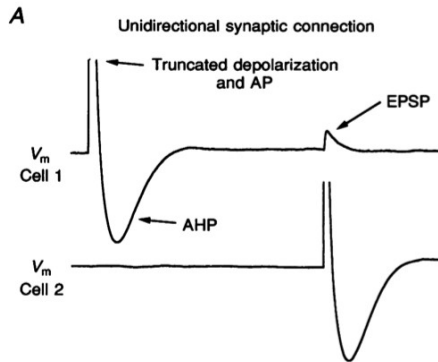
Inhibitory

- ▶ Make targets less likely to spike
- ▶ **GABA** (others)



Splettstoesser (2014) via Wikimedia

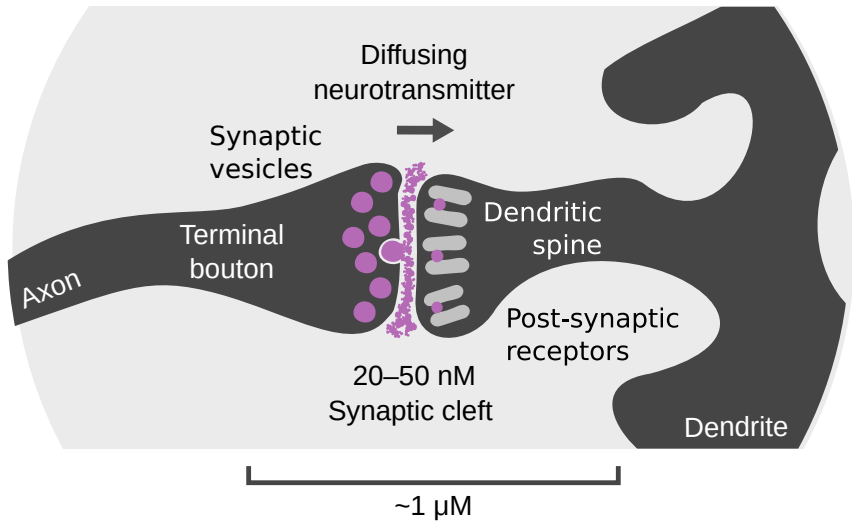
Chemical synapses are $\approx$ unidirectional



Markram et al, J Physiol (1997)

synapse $\rightarrow$ dendrite $\rightarrow$ soma $\rightarrow$ axon $\rightarrow$ synapse

Chemical synapses are $\approx$ unidirectional



draw a synapse, animate the steps of synaptic transmission

What is a synapse?

A connection between neurons that converts the action potential from a presynaptic neuron's axon (output) into a post-synaptic potential that acts as input for a another, postsynaptic neuron.

What kind of connection is a chemical synapse?

An electrochemical connections between neuronal cells.

- ▶ They can be excitatory, inhibitory (modulatory, other...)
- ▶ They are nonlinear, so can perform computations.
- ▶ They are plastic, so can store information (learning, memory)
- ▶ Neurotransmitter release is unreliable, so a synapse can be considered a probabilistic form of communication.

Steps in synaptic transmission

a *Presynaptic*

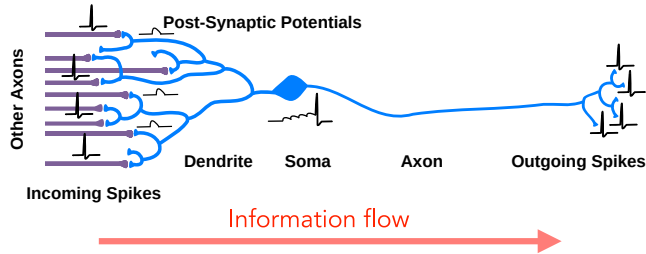
- 1 Action potential arrives
- 2 Voltage gated calcium channels open
- 3 Ca^{2+} enters the presynaptic cell
- 4 Ca^{2+} triggers vesicle fusion

b ((Neurotransmitter diffuses across synapse))

c *Postsynaptic*

- 1 Neurotransmitter binds to receptors on postsynaptic membrane
- 2 Ionotropic receptors open, allowing ionic currents
Metabotropic receptors trigger signalling cascades that modify ionic conductances
- 3 Ionic currents change dendritic voltage

ligand/voltage-gated, EPSP/IPSP



- 1 A pulse of neurotransmitters is released into synaptic cleft and binds to postsynaptic receptors
- 2 Receptors open, causing ionic currents that change the postsynaptic voltage ~ 1 mV for ~ 10 ms

↑ **Post-Synaptic Potential** PSP

- 3 Voltages sum at the soma; if $V_{\text{soma}} > V_{\text{threshold}}$, the cell spikes.
- 4 This spike travels out the axon triggering neurotransmitter release from axonal terminals

(repeat)



ligand/voltage-gated, EPSP/IPSP

end

... Quantum Chemistry, Molecular dynamics

Molecules

Gillespie, Master Equation

Concentrations

Mass-Action Kinetics

Conductance Models

Hodgkin-Huxley

Spiking Models

Leaky Integrate and Fire

Rate Neurons

Neural Mass/Field Models

Poisson Neurons

Generalized Linear Models

Binary Neurons

McCulloch-Pitts, Hopfield, Perceptron

Cognitive Neuroscience, Psychology ...

Physiological,
Quantitative

Biological
Realism,
Data needed
to identify
parameters



Computational
Efficiency,
Mathematical
Tractability



Phenomenological,
Qualitative